

Assignment 3

Day 2: Diagnosing Hardware failures Theory (30minits): common hardware issues and their system. Practical (30 min): Test Components using a multimeters.

Task 1: Five Common Hardware Issues

No	Hardware Issue	Typical Symptom
1	No Display	Monitor shows a blank/black screen and there may be no POST beep.
2	RAM Failure	PC may not boot, may restart repeatedly, or may produce beep codes
3	Hard Disk/SSD Failure	Operating system does not start or shows a boot/device error.
4	Power Supply Failure	PC does not turn on or suddenly shuts down.
5	CPU Overheating	PC becomes very slow, freezes, restarts, or shuts down unexpectedly

Task 2: Laptop Not Charging

Scenario

A laptop is not charging even when the charger is plugged in.

Troubleshooting Procedure

Step 1 – Check the power socket

Connect another electrical device to the same socket and check whether it is working.

Step 2 – Inspect the charger

Check the charger adapter, cable and connector for cuts, burns, loose connections or physical damage.

Step 3 – Check the laptop charging port

Inspect the charging port for dust, damage or looseness.

Step 4 – Check charger output voltage

Set the multimeter to **DC Voltage** and check the charger's output according to the voltage printed on the charger.

Step 5 – Check the battery connection

If the battery is accessible, check that its connector is properly connected.

Step 6 – Test with a compatible charger

If available, test the laptop with another compatible charger.

Conclusion:

If the charger and power socket are working but the laptop still does not charge, the problem may be related to the charging port, battery, charging circuit or motherboard.

Task 3: Testing USB Phone Charger Voltage

Objective

To measure the output voltage of a standard USB phone charger using a multimeter.

Tools Required

- Digital multimeter
- USB phone charger
- USB cable/USB connector

Procedure

1. Set the multimeter to **DC Voltage (V --)**.
2. Select a suitable voltage range, such as **20 V DC**.
3. Connect the charger to the power supply.
4. Place the **black probe on Ground/Negative**.
5. Place the **red probe on the +5 V contact**.
6. Observe and record the voltage shown on the multimeter.

Observation

Measured Voltage: 5.08 V DC

Expected Range

USB voltage should normally be approximately **4.75 V to 5.25 V** for a nominal 5 V supply.

Result : The measured voltage was 5.08 V, which is within the expected range.

Conclusion: The USB charger provided a normal 5 V output during the test.

Task 4: Testing PC Power Supply 12 V Rail

To test the 12 V output of an ATX power supply using a multimeter.

Tools Required

- Digital multimeter
- ATX power supply connector
- PC power supply

Procedure

1. Turn OFF the computer and disconnect it from the AC power.
2. Disconnect the ATX connector from the motherboard.
3. Set the multimeter to **DC Voltage (V=)**.
4. Select a suitable range, such as **20 V DC**.
5. Identify the **yellow wire (+12 V)**.
6. Identify a **black wire (Ground)**.
7. Connect the **black probe to the black wire**.
8. Connect the **red probe to the yellow wire**.
9. Observe the voltage displayed on the multimeter.
10. Record the measured voltage.

Observation : Measured voltage: 12.05 V DC

Result: The measured voltage is approximately 12 V, which indicates that the 12 V rail is providing a voltage close to its nominal value.

If voltage is below 11.5 V: A reading below 11.5 V can indicate a possible PSU/12 V rail problem. It may contribute to random shutdowns or system instability.

Conclusion: The 12 V rail of the power supply can be checked with a multimeter to help identify possible PSU-related problems.